V2X-MARL-BENCH

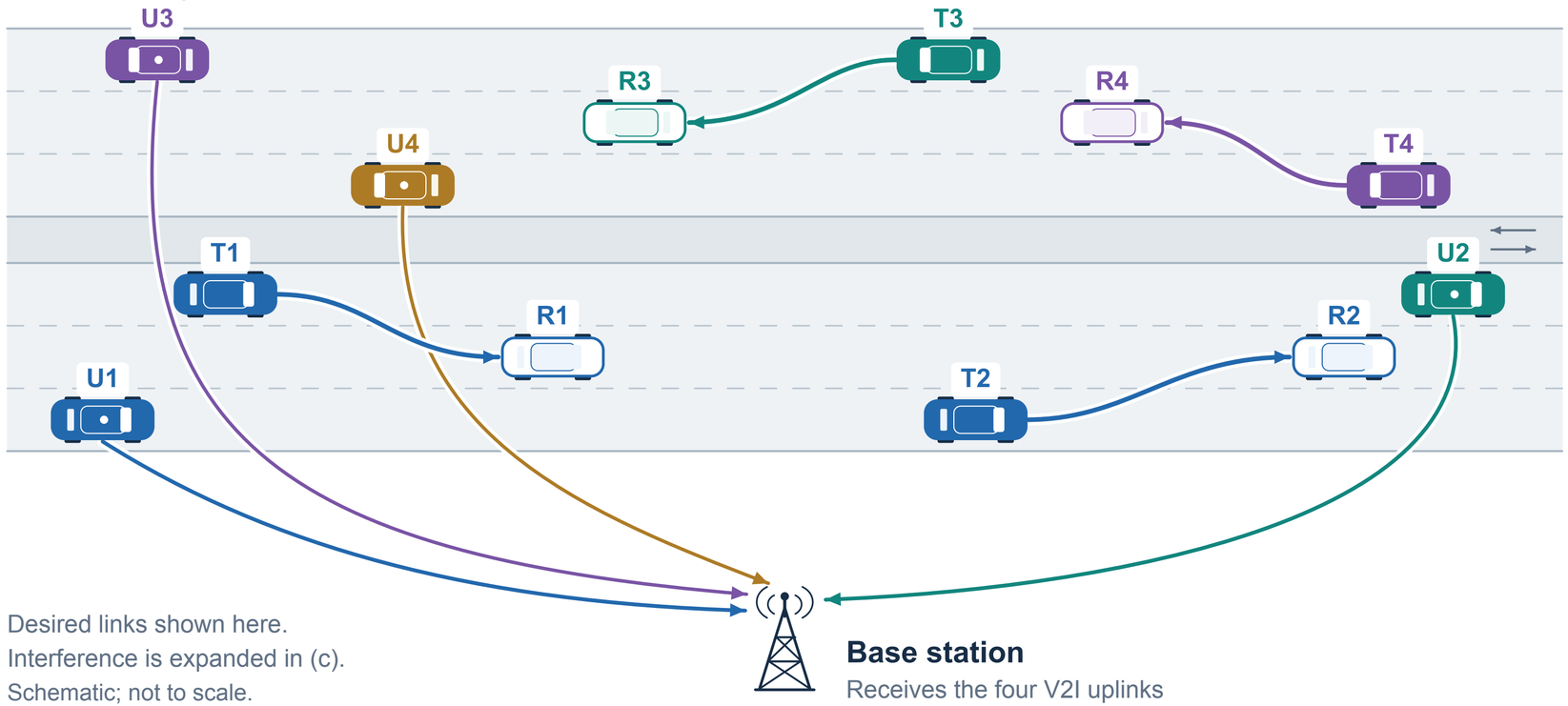
C-V2X sidelink resource allocation

SYSTEM MODEL

Shared uplink spectrum

(a) Highway deployment

Illustrative 4-agent allocation · three lanes in each direction



Desired links shown here.
Interference is expanded in (c).
Schematic; not to scale.

T: V2V transmitter / agent R: paired V2V receiver U: V2I user

(b) Spectrum reuse

Four orthogonal subchannels, 1 MHz each

SC 1 — **U1 → BS**
T1 → R1 T2 → R2

SC 2 — **U2 → BS**
T3 → R3

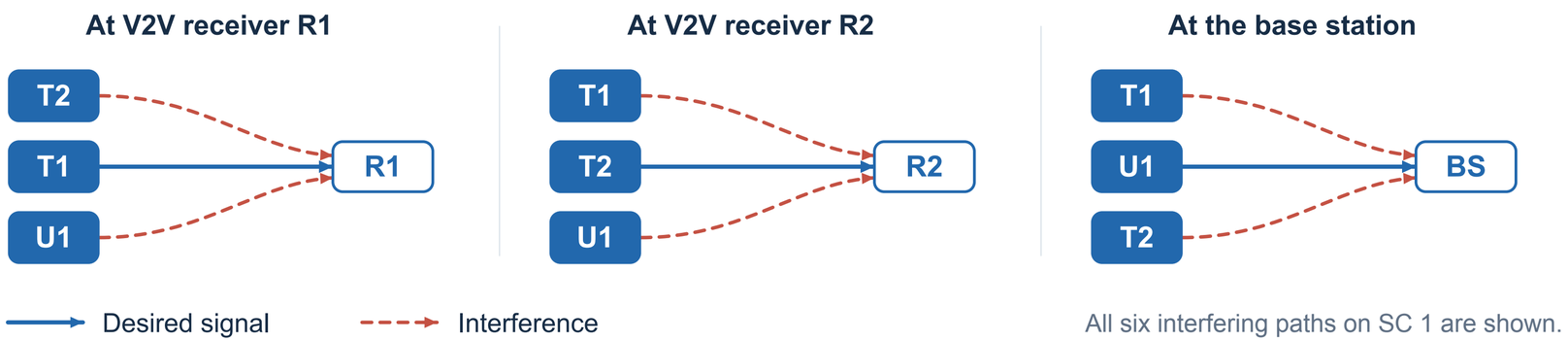
SC 3 — **U3 → BS**
T4 → R4

SC 4 — **U4 → BS**
No V2V reuse in this example

One fixed V2I uplink per subchannel.
V2V agents select which one to reuse.

(c) Co-channel interference

SC 1 expanded



All six interfering paths on SC 1 are shown.

(d) Agent decision

Each V2V transmitter selects

Subchannel	**SC 1, 2, 3 or 4**
Transmit power	**23, 10 or 5 dBm**

or chooses no transmission

4 × 3 + 1 = 13 actions

Benchmark: 4, 8 or 16 V2V agents · 4 V2I users · 1 ms decision steps in SIG / POSIG

Link colors identify subchannels.